IT342_Phase2_Mobile_CleanIT_Macansantos

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How Registration Works

- **Mobile Side:** The user fills in Full Name, Email, Password, and Contact Number, and selects a role (Client or Technician).
- **Request Sent:** The app uses RetrofitClient.instance.register() and sends a JSON body to the backend via POST /api/v1/auth/register.
- **Backend Side:** The Spring Boot backend receives the data, hashes the password using BCrypt, assigns the selected role (client or technician), and saves the user to your PostgreSQL database hosted on Supabase.
- **Response:** On success, the app automatically triggers a login with the same credentials and redirects to the Home screen. On failure, the app displays the error message as a Toast (e.g., "Email already exists").

How Login Works

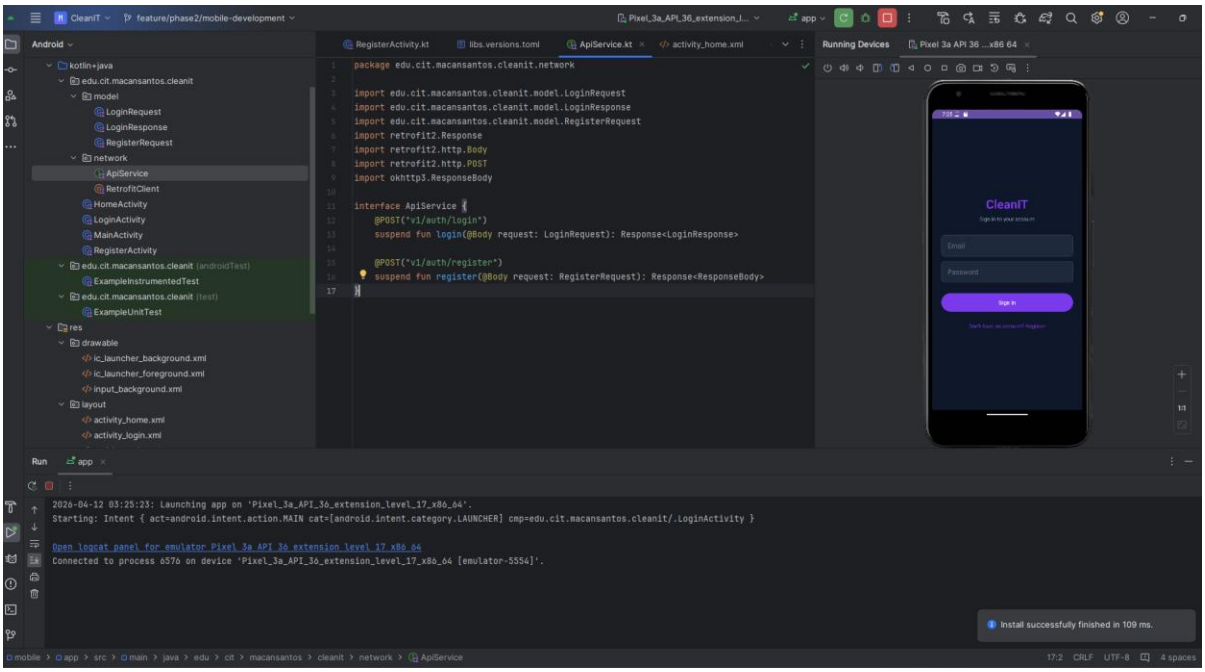
- **Mobile Side:** The user inputs their email and password.
- **Request Sent:** The app calls RetrofitClient.instance.login() and sends an HTTP POST request to /api/v1/auth/login.
- **Backend Side:** The Spring Boot backend queries the Supabase PostgreSQL database, verifies the email exists, and checks the password against the BCrypt hash. If valid, the server generates and returns a JWT token along with the user's name and role.
- **Response:** The app receives a LoginResponse containing the token, name, and role, then navigates to the Home screen displaying the user's name and role.

API Integration Used

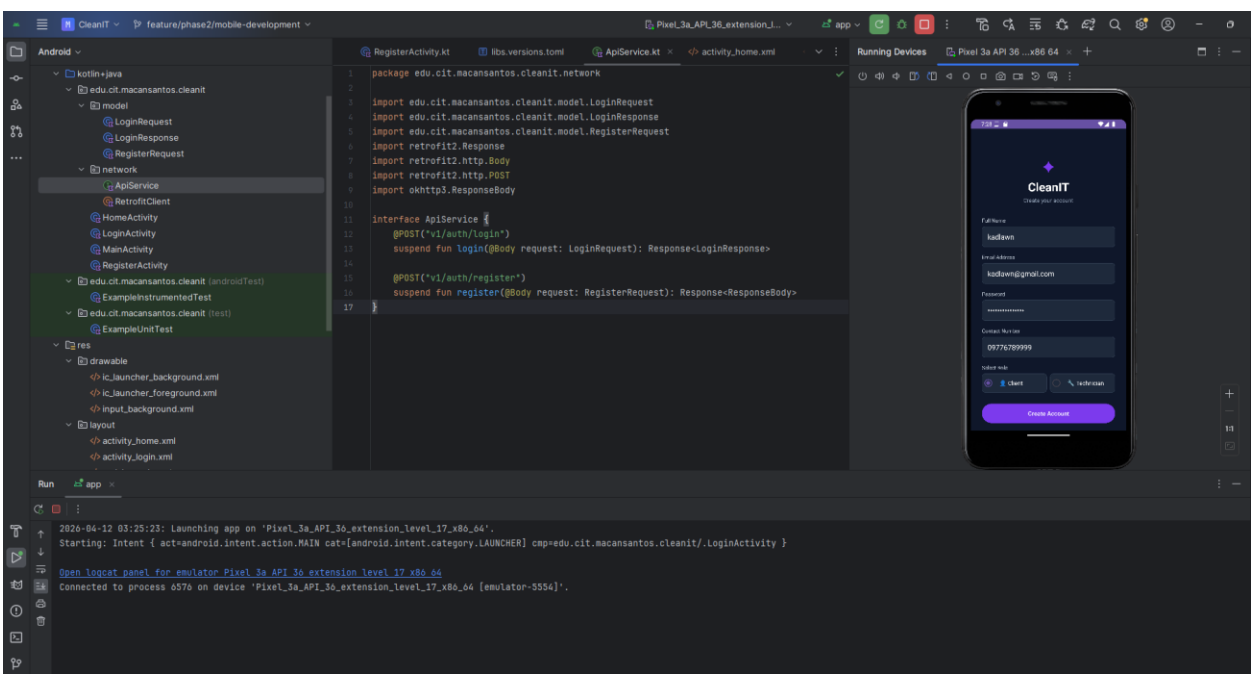
- **Library:** The mobile app uses Retrofit2 paired with Gson converter for serialization.
- **How it works:** When a network request is made, Retrofit constructs an exact HTTP POST call, converts Kotlin data classes (LoginRequest, RegisterRequest) into JSON using Gson, and sends them to the Spring Boot backend at BASE_URL = http://10.0.2.2:8080/api/ — the Android emulator's way of reaching localhost.

- **Concurrency:** All API calls use Kotlin Coroutines (lifecycleScope.launch with suspend fun), meaning network operations run off the main thread without blocking the UI, and results are safely delivered back to update the screen.

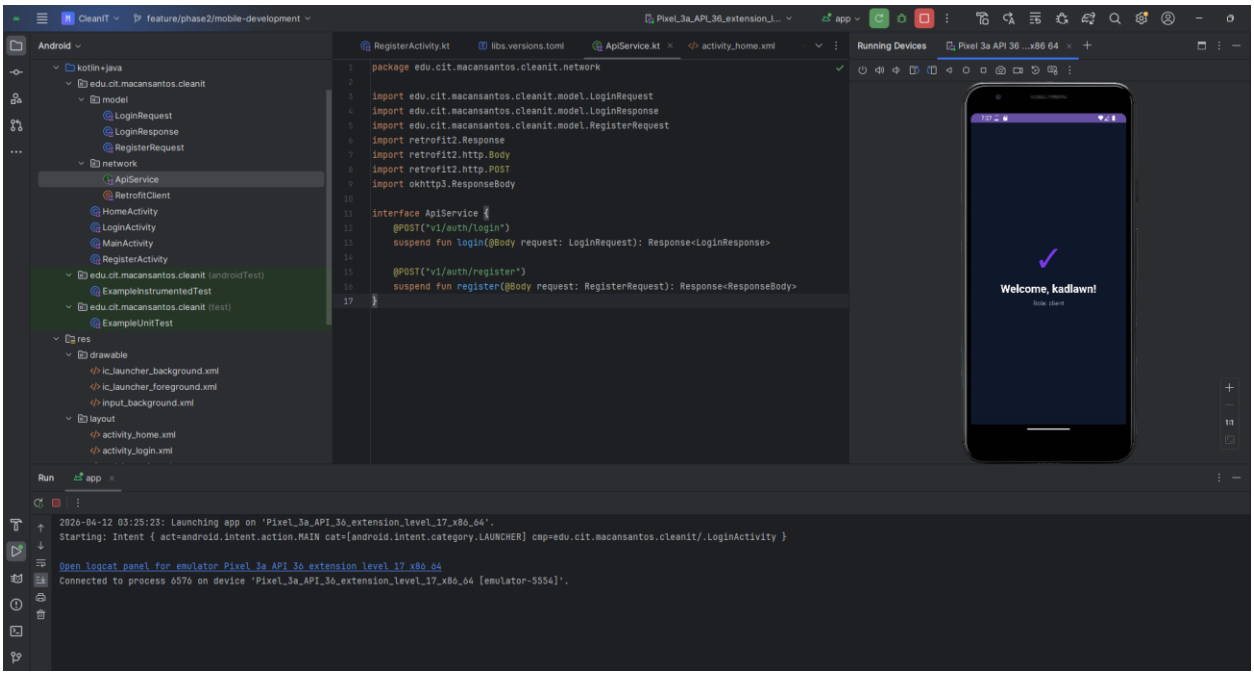
LOGIN SCREEN



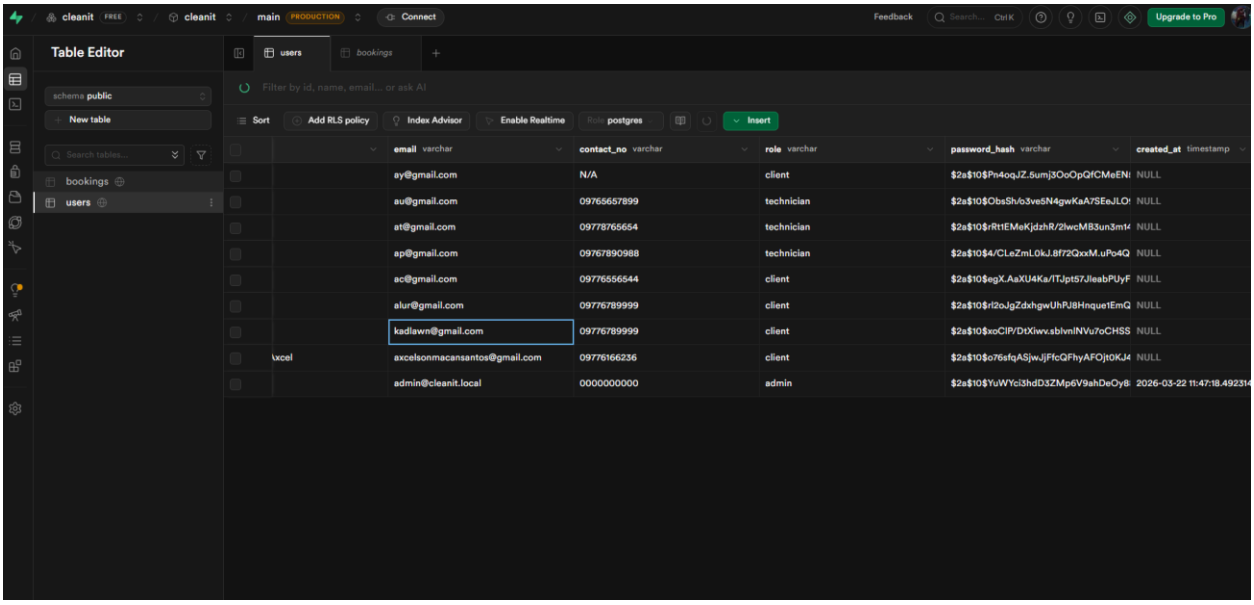
REGISTER SCREEN



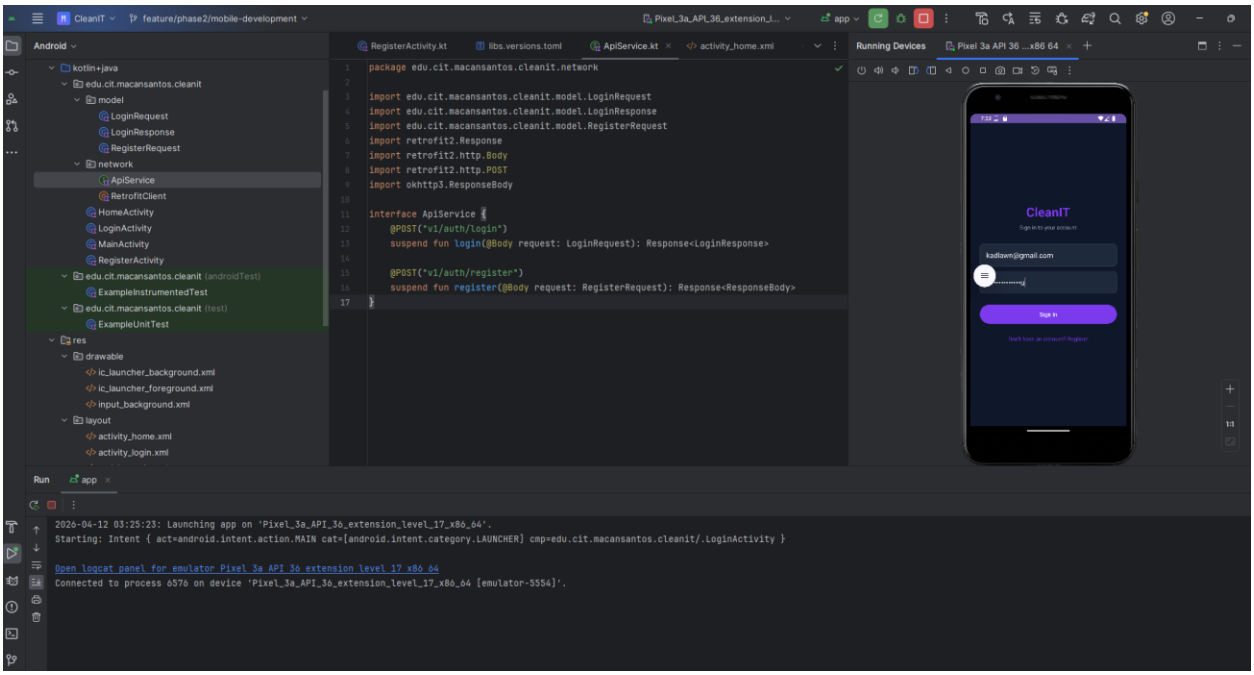
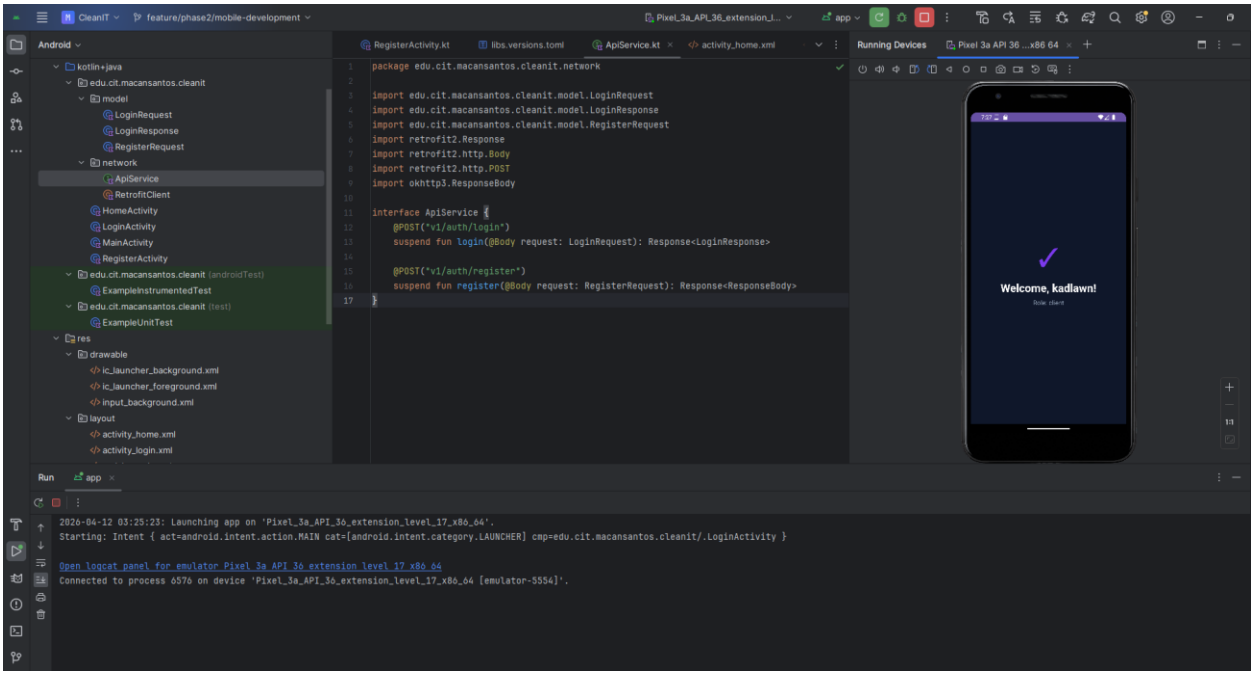
SUCCESSFUL REGISTER



SUPABASE / NEW ACCOUNT (kadlawn)



LOGIN WITH NEWLY MADE CREDENTIALS



SUCCESSFUL LOGIN SCREEN